



Technische Hochschule Deggendorf

**Case study project- EEG-Based Brain–Computer Interface
for Assistive Robotic Arm Control**

Bibliography Report

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Abstract

Approximately 15 million people worldwide experience a stroke each year, with around 5 million of those survivors sustaining permanent motor disability. For individuals affected by severe upper-limb motor impairments — whether caused by stroke, spinal cord injury, cerebral palsy, or neurodegenerative disease — Brain–Computer Interface (BCI) technology represents one of the most impactful pathways toward restored independence. By establishing a direct link between neural activity and external devices, BCIs remove the reliance on intact muscular pathways and enable command and control of assistive systems purely through cognitive intention.

This report presents a comprehensive bibliographic study and methodological outline for the development of an EEG-based BCI system designed to control a robotic arm in an assistive robotics context. Electroencephalography (EEG) is selected as the primary signal modality on account of its non-invasive nature, relatively low cost, and well-established role in motor imagery research. The study systematically reviews state-of-the-art techniques in EEG signal acquisition, preprocessing, spectral and spatial feature extraction, machine-learning-based intention classification, and BCI-to-robot interface design.

A structured, modular processing pipeline is proposed, comprising EEG data acquisition from publicly available datasets, band-pass filtering for artefact suppression, feature extraction using Power Spectral Density (PSD) and Common Spatial Patterns (CSP), classification using Support Vector Machines (SVM) or Logistic Regression, and output command mapping to a simulated robotic arm. The simulation-first development approach allows safe, cost-effective algorithmic validation prior to physical hardware integration.

The report further identifies critical research gaps, including over-reliance on laboratory-grade EEG hardware, limited modular system architectures, inadequate real-time latency optimisation, and insufficient simulation-based validation frameworks in the existing literature. The expected outcomes include a functional BCI processing pipeline, a trained classification model achieving at least 75% accuracy, and a simulated robotic arm capable of executing 4–5 basic directional commands with a system response latency not exceeding 500 milliseconds.

1. Introduction

1.1 Motivation

The convergence of artificial intelligence, robotics, and neurotechnology has created unprecedented opportunities for restoring quality of life to individuals living with physical disabilities. Among the most impactful of these developments is the Brain–Computer Interface — a technology that establishes a direct communication channel between the human brain and external devices, bypassing conventional neuromuscular pathways entirely. For individuals who have lost voluntary motor control due to stroke, traumatic spinal injury, amyotrophic lateral sclerosis (ALS), or cerebral palsy, BCIs offer a means of operating assistive devices using only cognitive intention.

Assistive robotic arms represent one of the most practically significant BCI applications. By decoding motor imagery from EEG signals and mapping the decoded intentions to robotic movements, it becomes possible for a person with severe upper-limb paralysis to reach, grasp, and manipulate objects in their environment. This capability can meaningfully restore independence in

activities of daily living, including eating, drinking, self-care, and interacting with household devices.

Despite considerable research progress, the majority of BCI-controlled robotic systems remain confined to controlled laboratory settings. They depend on expensive medical-grade EEG amplifiers, require extensive user calibration sessions, and rely on computationally intensive models that are difficult to deploy in real time on standard hardware. There is therefore a clear and pressing need for simpler, more scalable, and more accessible BCI systems — ones that can be practically implemented, fully validated in simulation, and eventually deployed without specialised infrastructure.

This project directly addresses that need by proposing a lightweight, modular, and simulation-validated EEG-BCI pipeline for assistive robotic arm control.

1.2 Objectives

The primary objectives of this study are as follows:

- To review and critically evaluate existing literature on EEG-based BCIs, motor imagery classification, and assistive robotic arm control
- To identify suitable signal preprocessing, feature extraction, and machine-learning techniques appropriate for a real-time, lightweight BCI system
- To identify research gaps that motivate a simplified, modular, and simulation-validated approach
- To propose and justify a structured technical pipeline for the proposed system
- To define measurable performance targets (KPIs) grounded in the reviewed literature

1.3 Scope of Study

This literature review covers publications from 2000–2025, with primary emphasis on work published after 2015. Sources include peer-reviewed journal articles, conference proceedings, and technical books from IEEE Xplore, PubMed, Springer, and Google Scholar. The review spans the following technical domains:

- Non-invasive EEG-based BCI systems and paradigms
- Motor imagery signal acquisition and preprocessing
- Spectral and spatial feature extraction methods
- Classification algorithms including traditional ML and deep learning
- BCI integration with assistive and rehabilitation robotic arms
- Simulation environments for BCI development and validation

2. Background and Fundamentals

2.1 Brain–Computer Interfaces

A Brain–Computer Interface is a system that acquires, analyses, and translates brain signals into commands for external output devices. BCIs create a direct channel between neural activity and a machine, bypassing the normal neuromuscular pathway through which the brain communicates movement intentions to the body.

BCIs are broadly categorised according to the invasiveness of signal acquisition. Invasive BCIs, such as intracortical microelectrode arrays, provide high spatial resolution but require surgical implantation, carrying associated risks. Semi-invasive methods, such as electrocorticography (ECoG), are placed on the brain surface beneath the skull. Non-invasive BCIs — most commonly using EEG — capture electrical potentials from the scalp without any surgical procedure. While non-invasive methods produce signals of lower spatial resolution and are more susceptible to noise, they are substantially safer, more affordable, and more widely accessible, making them the dominant modality in assistive and rehabilitative BCI applications.

2.2 Motor Imagery

Motor imagery (MI) refers to the mental simulation of a physical movement without any actual muscular activation. When a user imagines moving their left or right hand, or extending their arm upward, the brain generates distinct electrical patterns — particularly in the sensorimotor cortex — that differ measurably from resting-state baseline activity. These patterns are characterised by Event-Related Desynchronisation (ERD) and Event-Related Synchronisation (ERS) in the alpha (8–13 Hz) and beta (13–30 Hz) frequency bands.

Because these patterns are consistent, repeatable, and class-discriminative, motor imagery has become the most widely used paradigm in EEG-based BCIs for robotic control. With appropriate training, users can reliably generate distinct MI signals for several imagined movements, which are decoded by a classification algorithm and mapped to corresponding robot commands.

2.3 EEG Signal Characteristics and Frequency Bands

EEG captures voltage fluctuations resulting from the synchronised activity of populations of cortical neurons. Signals are extremely small — typically in the range of 10–100 microvolts at the scalp — and are corrupted by noise from eye blinks, muscle activity, electrical mains interference, and electrode movement.

EEG signals are characterised by their frequency content, which is divided into the following physiologically meaningful bands:

Table 1: EEG Frequency Bands and Associated States

Band	Frequency Range	Associated Physiological State
Delta	0.5 – 4 Hz	Deep sleep, unconsciousness
Theta	4 – 8 Hz	Drowsiness, memory encoding
Alpha	8 – 13 Hz	Relaxed wakefulness, motor imagery ERD
Beta	13 – 30 Hz	Active cognition, motor imagery ERS
Gamma	> 30 Hz	High-level cognitive processing

For motor imagery BCIs, the alpha and beta bands carry the most discriminative information relevant to imagined movement and are the primary targets for feature extraction and classification throughout this project.

2.4 BCI System Architecture

A standard EEG-based BCI system is composed of a sequential set of processing stages. Each stage transforms the signal from raw electrical activity into an actionable robot command.

Table 2: Standard BCI System Components

Component	Function	Methods Commonly Used
Signal Acquisition	Capture EEG from scalp electrodes	EEG headset, amplifier, dataset
Preprocessing	Remove noise and artefacts	Band-pass filtering, ICA, epoch rejection
Feature Extraction	Represent signal discriminatively	CSP, PSD, wavelet transform
Classification	Predict user intent class	SVM, LDA, CNN
Control Interface	Map prediction to robot command	Serial communication, ROS
Output Device	Execute physical or simulated action	Robotic arm, simulation environment

3. Literature Review

3.1 Foundations of EEG-Based Brain–Computer Interfaces

Wolpaw et al. (2002) published one of the most widely cited foundational studies in BCI research, demonstrating that EEG signals could be reliably used to communicate and control external devices in real time. Their work established that with appropriate training and signal processing, users could modulate sensorimotor rhythms in the alpha and beta frequency bands to generate stable and repeatable BCI control signals. This seminal contribution defined the conceptual and technical framework upon which the majority of subsequent motor imagery BCIs have been built.

Key Insight: Sensorimotor rhythm modulation through motor imagery is a reproducible and reliable paradigm for BCI-based assistive control.

Relevance to Proposed Project: Validates the selection of motor imagery as the control paradigm and the alpha/beta frequency bands as the primary feature source in the proposed pipeline.

Pfurtscheller and da Silva (1999) conducted a detailed investigation into Event-Related Desynchronisation and Event-Related Synchronisation patterns in EEG, confirming that distinct neural signatures are reliably generated for imagined movements of different body parts. The study identified the sensorimotor cortex as the primary region of interest and quantified the timing and spectral characteristics of MI-related EEG changes, providing the neurophysiological basis for MI-based BCI classification.

Key Insight: ERD and ERS in the alpha and beta bands provide neurophysiologically grounded, class-discriminative features for motor imagery BCIs.

Relevance: Supports the targeting of the 8–30 Hz frequency range during preprocessing and feature extraction.

Lebedev and Nicolelis (2006) provide a broad review of brain–machine interfaces, tracing progress from early animal studies to human neuroprosthetics. Their analysis identifies the key challenges of signal non-stationarity, user adaptation, and decoder generalisation that continue to limit BCI performance, while also highlighting the enormous clinical potential of BCI technology for restoring communication and motor function in severely disabled patients.

Key Insight: Non-stationarity of EEG and inter-session variability are fundamental challenges for BCI system reliability.

Relevance: Motivates the use of session-specific preprocessing and subject-dependent classifier training in the proposed system.

3.2 EEG Signal Processing Techniques

Nicolas-Alonso and Gomez-Gil (2012) provide a comprehensive review of EEG signal acquisition, preprocessing, and analysis methods used in BCI research. The authors systematically compare filtering strategies including Butterworth band-pass filters, Laplacian spatial filters, and Independent Component Analysis (ICA). They demonstrate that robust preprocessing — particularly the suppression of electro-oculographic (EOG) artefacts from eye movement and electromyographic (EMG) artefacts from muscle activity — is an essential prerequisite for achieving reliable classification accuracy. The review concludes that a 4th-order Butterworth band-pass filter in the 8–30 Hz range represents the most consistently effective preprocessing choice for motor imagery tasks.

Key Insight: Structured preprocessing, including band-pass filtering and artefact rejection, is a prerequisite for reliable motor imagery classification.

Relevance: Directly defines the preprocessing steps applied in the proposed pipeline.

Table 3: Comparison of EEG Preprocessing Methods

Method	Purpose	Strength	Limitation
Band-pass filtering	Retain MI-relevant frequencies	Simple, low-latency	May retain some artefacts
Notch filter	Remove mains interference (50/60 Hz)	Targeted removal	Narrow effect
ICA	Blind source separation	Removes EOG/EMG well	Computationally expensive
Epoch rejection	Remove corrupted segments	Removes gross artefacts	Reduces data volume
Normalisation	Standardise amplitude across channels	Reduces inter-session drift	May distort some features

Makeig et al. (2004) demonstrate the application of ICA to separate brain-generated EEG components from non-neural artefacts. Their work shows that ICA-based artefact removal significantly improves the signal-to-noise ratio of EEG recordings without discarding valid neural data. This is particularly important for motor imagery BCIs where eye movement artefacts frequently overlap in frequency with the alpha band of interest.

Key Insight: ICA-based artefact removal improves signal quality without sacrificing useful EEG data.

Relevance: Informs the artefact rejection step of the proposed preprocessing pipeline.

3.3 Feature Extraction Methods

Ramoser et al. (2000) introduced the Common Spatial Patterns (CSP) algorithm, which has since become the standard approach for spatial feature extraction in EEG-based motor imagery BCIs. CSP computes a set of spatial filters that simultaneously maximise the variance of one class while minimising the variance of the other, yielding features with high inter-class discriminability. Log-variance features derived from CSP-filtered signals have consistently demonstrated strong performance on binary MI classification tasks across a wide range of subjects and datasets.

Key Insight: CSP provides highly discriminative spatial features for binary motor imagery classification and is computationally tractable for real-time systems.

Relevance: CSP is selected as the primary feature extraction method for the proposed system.

Welch (1967), in the foundational work on spectral estimation, introduced the periodogram averaging technique that underlies Power Spectral Density (PSD) estimation. PSD computes the distribution of signal power across frequency bins and is widely used in EEG research as a frequency-domain feature. In MI BCIs, PSD features in the alpha and beta bands provide a computationally inexpensive complement to spatial methods such as CSP.

Key Insight: PSD provides a fast, interpretable, and robust frequency-domain feature particularly suited to constrained real-time environments.

Relevance: PSD will be evaluated as a lightweight alternative and complement to CSP in the proposed feature extraction module.

Table 4: Comparison of Feature Extraction Techniques

Method	Domain	Description	Strength	Limitation
PSD	Frequency	Power per frequency band	Simple, interpretable	Limited spatial resolution
CSP	Spatial	Variance-maximising filters	High accuracy for binary MI	Requires sufficient data
Wavelet Transform	Time-frequency	Multi-scale decomposition	Captures non-stationarity	Higher computational cost
Autoregressive (AR)	Parametric	Model-based spectral estimate	Low data requirement	Model order selection

3.4 Machine Learning for EEG Classification

Lotte et al. (2018) published an extensive comparative review of classification algorithms for EEG-based BCIs, covering a 10-year period of research. The analysis includes Linear Discriminant Analysis (LDA), Support Vector Machines (SVM), k-Nearest Neighbours, Naive Bayes, deep learning approaches including CNNs and RNNs, and Riemannian geometry methods. Their analysis shows that SVM consistently achieves high accuracy on motor imagery tasks with small training datasets, maintains good generalisation, and produces inference times compatible with real-time control. Riemannian geometry approaches perform comparably but require greater implementation effort.

Key Insight: SVM is robust, accurate, and computationally efficient for EEG classification — particularly suitable for resource-constrained real-time systems with limited training data.

Relevance: Directly justifies the selection of SVM as the primary classifier in the proposed project.

Roy et al. (2019) provide a systematic review of deep learning architectures applied to EEG analysis, comparing CNNs, RNNs, autoencoders, and hybrid models. While deep learning approaches achieve higher accuracy on large datasets, they require substantially greater computational resources, longer training times, larger training datasets, and careful hyperparameter tuning. The authors note that for real-time systems operating on standard hardware with limited data availability — conditions directly applicable to the proposed project — classical ML methods frequently outperform deep learning in practice.

Key Insight: Deep learning improves classification accuracy on large datasets but introduces computational overhead incompatible with low-latency real-time deployment in resource-constrained settings.

Relevance: Supports prioritising SVM and Logistic Regression over deep learning architectures in the proposed system.

Blankertz et al. (2007) demonstrate the feasibility of single-trial EEG classification, showing that meaningful BCI commands can be decoded from a single, brief EEG epoch — typically 1–4 seconds in duration — without the need for time-consuming trial averaging. This is fundamental to real-time BCI systems, which must respond to user intent within a few hundred milliseconds to maintain natural interaction.

Key Insight: Single-trial classification is achievable with well-designed features and trained classifiers, enabling real-time BCI control within latency constraints.

Relevance: Supports the project KPI of response latency ≤ 500 ms.

Table 5: Comparison of Classification Algorithms for EEG BCIs

Algorithm	Type	Typical Accuracy (MI)	Computational Cost	Real-time Suitability
SVM (RBF kernel)	Classical ML	75–85%	Low	Excellent
Logistic Regression	Classical ML	70–80%	Very Low	Excellent
LDA	Classical ML	70–78%	Very Low	Excellent
CNN	Deep Learning	80–90%	High	Moderate
RNN / LSTM	Deep Learning	78–88%	Very High	Poor
Riemannian Geometry	Geometric ML	80–87%	Moderate	Good

3.5 BCI-Controlled Robotic Arm Systems

Abiri et al. (2019) present a systematic review of BCI-controlled robotic and prosthetic devices spanning more than 200 studies. Their analysis identifies the most effective BCI paradigms for robotic arm control, the range of achievable command sets — typically 2–6 discrete commands — and the hardware configurations most suitable for practical assistive use. A key finding is that system simplicity, low user training burden, and command execution reliability are the primary

determinants of real-world usability, significantly outweighing marginal gains in raw classification accuracy.

Key Insight: Effective assistive BCI systems prioritise usability, reliability, and a small but sufficient command set over maximum classification performance.

Relevance: Confirms the project decision to target 4–5 basic robotic arm commands and to prioritise real-time latency over classifier sophistication.

Padfield et al. (2019) review EEG-based BCI control systems with an emphasis on motor imagery paradigms and their practical application to robotic devices. The study demonstrates the feasibility of operating a robotic arm with three to five motor imagery classes and highlights the critical importance of subject-specific calibration to account for inter-individual EEG variability. The authors further identify that a consistent preprocessing pipeline is the single most influential factor in maintaining classification stability across sessions.

Key Insight: User-specific calibration and a consistent preprocessing protocol are the most important practical factors for MI-based robotic BCI reliability.

Relevance: Informs the calibration strategy and motivates session-normalised preprocessing in the proposed system.

Dávila-Chacón et al. (2018) develop a low-cost, multi-joint robotic arm platform designed explicitly for BCI integration in assistive and rehabilitation contexts. Their system uses a modular hardware design paired with a Python-based control interface, achieving reliable basic grasping and directional movements under BCI-driven control. The work demonstrates that functional assistive robotic arm control does not require expensive commercial platforms such as the KUKA or Jaco arms that dominate laboratory settings.

Key Insight: A low-cost, modular robotic platform is sufficient for assistive BCI arm control, making the technology accessible for academic and clinical development.

Relevance: Supports the project approach of using a simulation or low-cost physical arm prototype, validating affordability as a design constraint.

Table 6: Summary of BCI-Controlled Robotic Arm Studies

Study	BCI Paradigm	Command Set	Classifier	Key Achievement
Abiri et al. (2019)	Motor imagery	2–6 commands	SVM, LDA	Systematic usability review
Padfield et al. (2019)	Motor imagery	3–5 commands	SVM	Subject-specific calibration
Dávila-Chacón et al. (2018)	Mixed BCI	4 commands	SVM	Low-cost arm platform
Wolpaw et al. (2002)	Sensorimotor rhythms	2 commands	LDA	Foundational BCI control
Blankertz et al. (2007)	Motor imagery	2–4 commands	LDA/SVM	Single-trial real-time control

3.6 Simulation in BCI Development and Validation

Lotte et al. (2013) argue that simulation-first development is a recognised best practice in BCI research, enabling complete algorithmic validation without the operational risks associated with direct hardware control during early-stage development. Their work describes a structured pipeline in which signal processing and classification models are fully verified within a simulated control environment before integration with physical robotic systems. This approach reduces the risk of hardware damage, user safety incidents, and wasted development time caused by undetected algorithmic errors.

Key Insight: Simulation-based validation reduces development risk, accelerates iteration, and enables reproducible benchmarking prior to physical hardware integration.

Relevance: Directly justifies the simulation-first approach of the proposed project using PyBullet or Gazebo.

Côté-Allard et al. (2019) demonstrate that transfer learning combined with simulation environments significantly reduces the dataset requirements for effective BCI classifier training. Their work shows that models trained offline on publicly available EEG datasets, and validated in simulation, generalise substantially better to new subjects than models trained on small proprietary datasets alone. This finding is particularly significant for academic BCI projects where access to large pools of human participants is limited.

Key Insight: Publicly available EEG datasets combined with simulation environments provide a reproducible and scalable foundation for BCI development.

Relevance: Supports the project plan to use publicly available datasets — BCI Competition IV Dataset 2a and PhysioNet EEG Motor Movement/Imagery Dataset — for training and simulation-based evaluation.

4. Comparative Analysis of Reviewed Studies

The reviewed literature reveals three dominant methodological approaches for EEG-based BCI robotic arm control. A structured comparison of representative works across these approaches is presented below.

Table 7: Comparative Analysis of Reviewed Approaches

Approach	Study	Feature Method	Classifier	Command Set	Performance	Limitation
Classical ML pipeline	Blankertz et al. (2007)	CSP	LDA	2–4	~80% accuracy	Binary or limited classes
SVM-based pipeline	Lotte et al. (2018)	CSP + PSD	SVM	3–5	75–85%	Parameter tuning
Deep learning pipeline	Roy et al. (2019)	CNN features	CNN	4–6	80–90%	High data and compute cost
Hybrid approach	Padfield et al. (2019)	PSD + CSP	SVM	3–5	~78%	User variability

Approach	Study	Feature Method	Classifier	Command Set	Performance	Limitation
Low-cost robotic arm	Dávila-Chacón et al. (2018)	PSD	SVM	4	Functional	Hardware dependent
Simulation-validated	Lotte et al. (2013)	CSP	LDA/SVM	2–4	Reproducible	Simulation gap

The comparison demonstrates that classical ML pipelines using CSP or PSD features with SVM classifiers consistently achieve competitive performance across a wide range of conditions, while remaining computationally tractable for real-time deployment. Deep learning approaches offer performance advantages on large datasets but introduce deployment barriers that are inconsistent with the objectives and constraints of the proposed project.

5. Summary Table of All Reviewed Works

Table 8: Summary of All Reviewed Literature

Reference	Focus Area	Method	Key Contribution	Relevance
Wolpaw et al. (2002)	BCI foundations	Sensorimotor rhythms	BCI framework for communication and control	Paradigm selection
Pfurtscheller & da Silva (1999)	Motor imagery	ERD/ERS analysis	Neural basis of MI-EEG patterns	Feature band justification
Lebedev & Nicolelis (2006)	Brain–machine interfaces	Review	Historical and clinical context	Challenges identification
Nicolas-Alonso & Gomez-Gil (2012)	Preprocessing	Filtering, ICA	Comprehensive preprocessing guide	Pipeline design
Ramoser et al. (2000)	Feature extraction	CSP	Spatial filtering standard	Feature method selection
Makeig et al. (2004)	Artefact removal	ICA	Improved signal quality	Preprocessing step
Welch (1967)	Spectral estimation	PSD	Frequency-domain feature baseline	Feature method selection
Lotte et al. (2018)	Classification	SVM, LDA, CNN review	10-year algorithm benchmarking	Classifier justification
Roy et al. (2019)	Deep learning EEG	CNN, RNN, hybrid	Deep vs. classical ML trade-offs	Justifies avoiding deep learning
Blankertz et al. (2007)	Single-trial BCI	CSP + LDA	Real-time single-trial feasibility	Latency KPI support
Abiri et al. (2019)	BCI robotic review	Multiple	Usability and command reliability	Command set design
Padfield et al. (2019)	EEG robotic arm	CSP + SVM	Subject calibration importance	Training strategy
Dávila-Chacón et al. (2018)	Low-cost arm	PSD + SVM	Affordable BCI robotic platform	Hardware approach
Lotte et al. (2013)	Simulation BCI	CSP + LDA	Simulation as best practice	Simulation-first validation
Côté-Allard et al. (2019)	Transfer learning	CNN + transfer	Public datasets + simulation	Dataset and generalisation

6. Synthesis and Research Gaps

6.1 Synthesis of Findings

The reviewed literature reveals a clear and consistent trajectory in EEG-based BCI development. Foundational studies established that motor imagery reliably generates classifiable neural patterns in the alpha and beta frequency bands. Subsequent research refined preprocessing pipelines, culminating in robust artefact removal and spatial filtering methods — most significantly CSP — that substantially improve signal discriminability. The emergence of machine-learning-based classifiers, particularly SVM, brought real-time feasibility to EEG-based systems. More recent deep learning approaches have extended accuracy limits but at computational costs that are difficult to justify for constrained real-world assistive deployments.

In the domain of robotic arm control, the literature consistently demonstrates that 4–6 discrete motor imagery commands are sufficient for meaningful assistive tasks. Studies confirm that system simplicity, low latency, and reliable command execution are valued more highly by end users than marginal accuracy improvements. Simulation-based validation has emerged as a best-practice development strategy, enabling reproducible testing and de-risking the transition to physical hardware.

Three overarching design principles emerge from the synthesis:

- Simplicity — lightweight preprocessing and classical ML over computationally heavy deep architectures
- Modularity — independently testable pipeline stages that can evolve without full system redesign
- Simulation-first validation — complete algorithmic verification in simulation before physical hardware integration

6.2 Identified Research Gaps

Despite the substantial progress documented in the reviewed literature, five important gaps remain that directly motivate the design of the proposed system:

Gap 1 — Over-reliance on laboratory-grade hardware: The majority of BCI robotic studies depend on medical-grade EEG amplifiers costing thousands of euros and requiring expert operation. Very few studies demonstrate functional systems using affordable consumer-grade devices or publicly available datasets in an accessible academic setting.

Gap 2 — Insufficient simulation-based validation: While simulation is acknowledged as best practice, a significant proportion of published systems proceed directly to hardware testing without a documented and reproducible simulation validation stage, creating barriers for researchers without physical laboratory infrastructure.

Gap 3 — Lack of modular and scalable system architectures: Most published BCI systems are tightly coupled — preprocessing, classification, and control layers are not independently replaceable. This limits adaptability and makes individual component upgrades difficult.

Gap 4 — Inadequate real-time latency reporting: Many studies report offline classification accuracy without documenting end-to-end system latency, leaving it unclear whether the full pipeline could operate within the ≤ 500 ms threshold required for natural, usable robotic control.

Gap 5 — Limited multi-class command sets: The majority of EEG-BCI robotic arm studies demonstrate only two to three reliable motor imagery commands. This restricts practical utility for performing the variety of manipulation tasks required in daily living scenarios.

7. Alignment with the Proposed Work

The table below explicitly maps each key research insight and identified gap to the corresponding design decision in the proposed system.

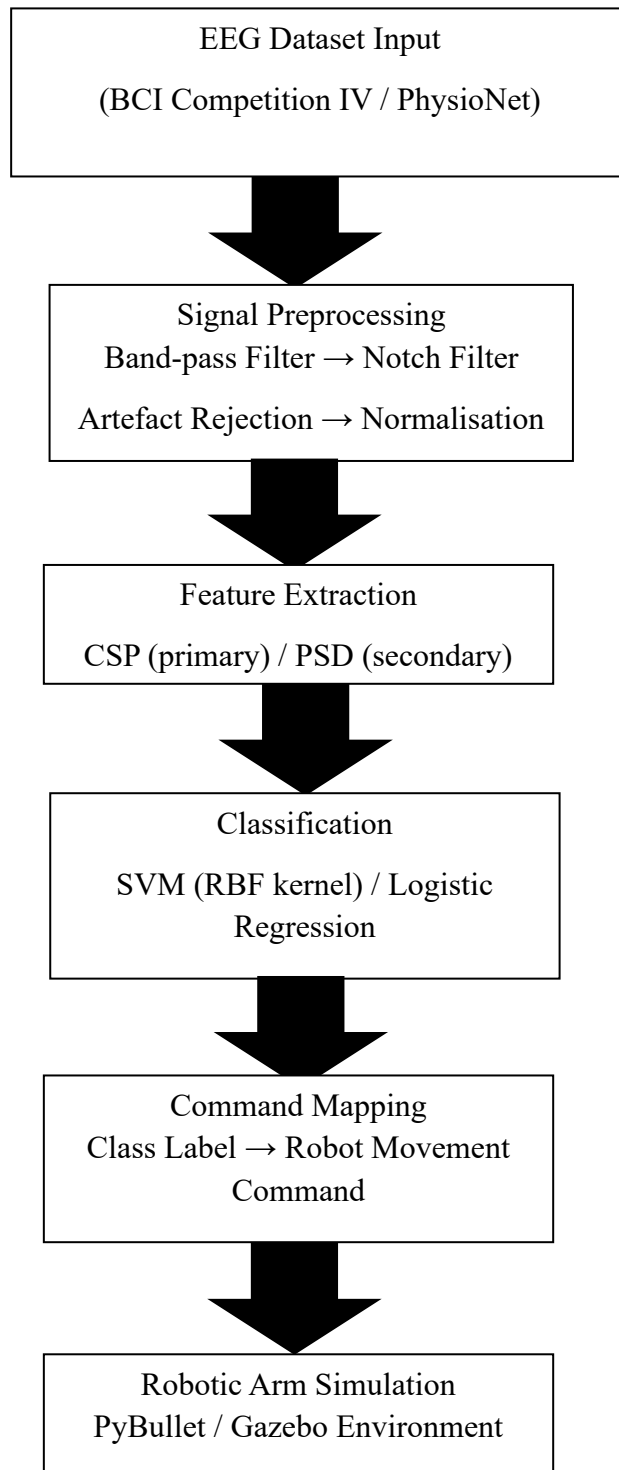
Table 9: Alignment of Literature Insights with Proposed System Design

Research Insight / Gap	Source	Proposed System Response
SVM competitive with deep models on small datasets	Lotte et al. (2018); Roy et al. (2019)	SVM selected as primary classifier
Band-pass filtering (8–30 Hz) essential for MI	Nicolas-Alonso & Gomez-Gil (2012)	Butterworth band-pass filter applied
CSP provides strongest spatial MI features	Ramoser et al. (2000)	CSP selected as primary feature method
Simulation reduces development risk and cost	Lotte et al. (2013)	PyBullet / Gazebo simulation stage
Tightly coupled architectures limit adaptability	Gap 3	Modular, independently testable pipeline
4–6 commands sufficient for assistive use	Abiri et al. (2019)	System targets 4–5 directional commands
Latency ≤ 500 ms required for natural BCI control	Blankertz et al. (2007)	Pipeline benchmarked against 500 ms KPI
Hardware dependency limits accessibility	Gap 1	BCI Competition IV / PhysioNet datasets used
Single-trial classification enables real-time BCI	Blankertz et al. (2007)	Single-trial epochs processed in pipeline

8. Proposed Methodology

8.1 System Architecture Overview

The proposed system is structured as a sequential five-stage modular pipeline. Each stage is independently implemented, tested, and replaceable, forming an architecture that directly addresses the modularity and scalability gaps identified in the literature.



8.2 EEG Data Acquisition

Rather than relying on physical EEG hardware during the development phase — which would introduce the hardware dependency gap identified in the literature — the system uses two publicly available, well-validated EEG datasets:

- BCI Competition IV Dataset 2a: Four-class motor imagery (left hand, right hand, feet, tongue), 22 EEG channels, 9 subjects, sampled at 250 Hz. This is the most widely used MI benchmark dataset in the literature and allows direct comparison with published results.
- PhysioNet EEG Motor Movement/Imagery Dataset: 64-channel EEG recordings from 109 subjects performing 14 motor and motor imagery tasks, sampled at 160 Hz. This dataset provides greater subject diversity for cross-subject generalisation evaluation.

Both datasets provide labelled EEG epochs corresponding to distinct motor imagery conditions, enabling supervised classifier training and rigorous cross-validation. Use of standardised datasets ensures full reproducibility of results.

8.3 Signal Preprocessing

Raw EEG signals contain substantial noise from multiple sources. The preprocessing stage applies a structured sequence of operations to improve signal quality:

1. Band-pass filtering — A 4th-order Butterworth filter with a pass-band of 8–30 Hz retains the alpha and beta components most relevant to motor imagery while suppressing low-frequency movement artefacts and high-frequency muscle noise.
2. Notch filtering — A 50 Hz notch filter removes mains electrical interference from the signal.
3. Artefact epoch rejection — Epochs in which the peak-to-peak amplitude of any channel exceeds $\pm 100 \mu\text{V}$ are discarded to eliminate gross movement or blink artefacts that cannot be removed by frequency filtering alone.
4. Epoch segmentation — Continuous EEG recordings are segmented into fixed-length windows of 2–4 seconds duration, time-locked to motor imagery cue onset.
5. Per-channel normalisation — Each epoch is normalised to zero mean and unit variance per channel to reduce the effect of inter-session and inter-subject amplitude drift.

8.4 Feature Extraction

Two complementary feature extraction approaches are evaluated:

Common Spatial Patterns (CSP): CSP computes a set of linear spatial filters that maximise variance for one class while simultaneously minimising it for the other. For the multi-class scenario, a one-versus-rest CSP decomposition is applied. Log-variance features are extracted from the spatially filtered signal for each filter pair. CSP has been shown across the reviewed literature to be the single most effective feature extraction method for motor imagery BCIs and is adopted as the primary method.

Power Spectral Density (PSD): PSD estimates are computed using Welch's method over the alpha (8–13 Hz) and beta (13–30 Hz) bands for a subset of selected electrodes over the motor cortex (C3, Cz, C4). PSD features provide a computationally inexpensive, interpretable complement to CSP and are evaluated as a secondary and potentially combined feature source.

Both methods are evaluated individually and in combination through cross-validated experiments, with feature selection based on classification accuracy.

8.5 Classification

Based on the literature review, two classification models are selected:

Support Vector Machine (SVM): SVM with a Radial Basis Function (RBF) kernel is trained on subject-specific data. Hyperparameters — regularisation parameter C and kernel width γ — are optimised through a nested 5-fold cross-validation grid search. SVM is selected as the primary classifier based on its consistently strong performance on small EEG datasets as reported by Lotte et al. (2018).

Logistic Regression: As a computationally simpler and fully interpretable baseline, Logistic Regression with L2 regularisation is evaluated for one-versus-rest multi-class scenarios. Its minimal inference time makes it the preferred option if the 500 ms latency KPI cannot be met with SVM.

Evaluation Protocol:

- 5-fold stratified cross-validation for model selection and hyperparameter tuning
- Subject-dependent evaluation (train and test on the same subject) for primary performance reporting
- Leave-one-subject-out cross-validation for cross-subject generalisation
- Primary metric: classification accuracy; secondary metrics: Cohen's Kappa and macro F1-score

8.6 Command Mapping and Robotic Arm Control

Classifier output labels are mapped to a fixed set of robotic arm commands. The proposed command set for 4–5 class control is:

Table 10: Motor Imagery to Robot Command Mapping

Class Label	Motor Imagery Task	Robot Command
Class 1	Left hand imagery	Move arm left
Class 2	Right hand imagery	Move arm right
Class 3	Both hands imagery	Move arm forward / initiate grasp
Class 4	Feet imagery	Move arm downward / release
Class 5	Rest / neutral	Hold current position

The command mapping module receives the predicted class label together with the classifier's posterior probability. A confidence threshold is applied — commands are only executed when the posterior probability exceeds 0.65 — to suppress false activations resulting from ambiguous EEG epochs. This mechanism directly addresses the real-time reliability concern identified in the literature.

8.7 Simulation Environment

The robotic arm is simulated using PyBullet, an open-source Python-based physics simulation library widely used for robotic research and prototyping. A 4-DOF robotic arm model is configured within the PyBullet environment to receive command signals from the classification and mapping modules and execute the corresponding joint movements in real time.

The simulation environment serves three distinct and critical purposes within the development workflow:

1. End-to-end pipeline validation: The complete pipeline — from EEG epoch input through preprocessing, feature extraction, classification, and command dispatch — is executed and verified within the simulation before any physical hardware is involved. This confirms that all modules communicate correctly and that the data flow is free of errors.
2. Latency benchmarking: The simulation provides a controlled environment for measuring the end-to-end system response time — defined as the elapsed time from EEG epoch onset to the initiation of the corresponding robot movement. This measurement is compared against the 500 ms KPI and used to identify and resolve pipeline bottlenecks.
3. Safety and robustness testing: The simulation allows testing of unintended command sequences, ambiguous classification outputs, and edge-case EEG epochs without any physical risk to hardware or users. Confidence thresholding and command smoothing strategies are evaluated and refined in simulation before hardware deployment.

The choice of PyBullet over alternatives such as Gazebo is motivated by its lightweight Python API, rapid setup time, and compatibility with the NumPy and scikit-learn libraries used in the upstream pipeline stages. A visual rendering window provides real-time feedback on arm movement during testing, supporting intuitive debugging.

9. Expected Outcomes and Performance KPIs

The following table defines the complete set of expected deliverables and measurable performance targets for the proposed system. Each KPI is grounded in and consistent with results reported in the reviewed literature.

Table 11: Expected Outcomes and Key Performance Indicators

Deliverable	Description	KPI / Success Criterion	Literature Basis
EEG preprocessing pipeline	Band-pass filter, notch filter, artefact rejection, normalisation	Measurable SNR improvement over raw signal	Nicolas-Alonso & Gomez-Gil (2012)
Feature extraction module	CSP and PSD feature computation	Statistically significant inter-class separation	Ramoser et al. (2000)
Trained classification model	SVM or Logistic Regression on MI dataset	$\geq 75\%$ classification accuracy	Lotte et al. (2018)
Command mapping interface	Class label to robot movement translation	$\geq 80\%$ command execution success rate	Abiri et al. (2019)
Simulated robotic arm control	4–5 arm movements under BCI pipeline control	All commands reliably and repeatably demonstrated	Dávila-Chacón et al. (2018)
End-to-end latency	Time from EEG epoch onset to robot response	≤ 500 ms response latency	Blankertz et al. (2007)
Reproducible codebase	GitHub repository with documented pipeline	Independently reproducible by third party	Côté-Allard et al. (2019)

10. Discussion

The proposed system design is grounded in a systematic and critical reading of the reviewed literature, with every architectural decision traceable to a specific research finding or identified gap. This section reflects on the most significant design choices, their justifications, and the challenges that will need to be managed during implementation.

Choice of SVM over Deep Learning: The decision to select SVM as the primary classifier, rather than more recent deep learning architectures such as CNNs or LSTMs, is one of the most consequential design choices in the proposed system. This decision is supported by Lotte et al. (2018) and Roy et al. (2019), both of whom demonstrate that SVM consistently achieves competitive accuracy on small, subject-specific EEG datasets while offering substantially lower computational overhead, shorter training times, and faster inference. Given that the proposed project operates on standard academic hardware without GPU infrastructure, and that the available training data per subject is limited to a few hundred trials per dataset, SVM represents the most pragmatically appropriate choice. Deep learning approaches may be revisited in future iterations when larger datasets and greater computational resources are available.

Simulation-First Development: The simulation-first approach directly addresses Gap 2 identified in the literature review — the absence of reproducible, documented simulation validation in many published BCI systems. By fully validating the end-to-end pipeline in PyBullet before any physical hardware is integrated, the proposed project ensures that algorithmic errors are caught and corrected at a stage where iteration is rapid and risk-free. This approach also produces a fully documented and reproducible validation baseline that is independently verifiable, which is a requirement of sound academic practice.

Modular Pipeline Architecture: The decision to implement the preprocessing, feature extraction, classification, and control modules as independently testable and replaceable units directly responds to Gap 3 — the tightly coupled, non-modular architectures prevalent in the existing literature. Modularity ensures that the system can evolve: for example, replacing the CSP feature extractor with a wavelet-based method, or upgrading the SVM classifier to a lightweight neural network, without requiring a complete system redesign. This positions the codebase as a genuinely reusable research artefact rather than a single-use implementation.

Inter-Subject Variability: One of the most significant practical challenges in EEG-based BCI development is inter-subject variability. EEG motor imagery patterns differ substantially between individuals — in terms of electrode distribution, frequency characteristics, and signal amplitude — and classifiers trained on one subject's data frequently perform poorly on another. The proposed system addresses this through subject-specific training and calibration, as recommended by Padfield et al. (2019). Future extensions may incorporate transfer learning techniques as described by Côté-Allard et al. (2019) to reduce per-subject calibration burden, enabling faster onboarding of new users.

Real-Time Latency: Meeting the 500 ms end-to-end latency KPI is a genuine engineering challenge. This threshold is physiologically motivated — delays longer than approximately 500 ms break the natural feedback loop between cognitive intention and observed robot response, reducing user confidence and control performance. In the proposed system, latency is managed through a

combination of lightweight feature extraction (CSP log-variance, band-power PSD), fast classifiers (SVM with precomputed kernel, Logistic Regression), and streamlined Python-to-simulation communication. The full latency profile will be measured and reported during the simulation validation phase, with optimisation applied to any stage found to be a bottleneck.

Ethical Considerations: Assistive BCI systems operate in a context that directly affects vulnerable users. System failures — whether spurious false commands or missed intended commands — can cause frustration, erode user trust, and in physical hardware contexts, pose safety risks. The proposed confidence thresholding mechanism, which suppresses commands with posterior probability below 0.65, represents a first-level safety measure. All user testing activities will comply with applicable ethical frameworks for human-subjects research. The simulation-first approach also means that no users are exposed to unvalidated hardware control at any stage of the project.

Transition to Physical Hardware: While the current project phase is focused on simulation-based development, the longer-term objective is integration with a physical robotic arm and, eventually, a consumer-grade EEG headset. The modular architecture ensures that this transition can be staged carefully — substituting the PyBullet simulation with a physical arm controller, and the dataset EEG input with a live signal acquisition interface — without restructuring the core processing pipeline. This planned trajectory from simulation to hardware is directly informed by the best-practice guidance of Lotte et al. (2013).

11. Conclusion

This report has presented a comprehensive bibliographic study and methodological framework for the development of an EEG-based Brain–Computer Interface for assistive robotic arm control. The literature review has confirmed that EEG motor imagery is a well-established and reliable BCI paradigm, that classical machine learning methods — particularly SVM with CSP features — are effective and computationally practical for real-time assistive systems, and that simulation-first development is both achievable and strongly recommended as a validation strategy for academic and clinical BCI projects.

Five critical gaps in the existing literature were systematically identified: over-reliance on expensive laboratory-grade hardware, insufficient simulation-based validation, lack of modular and scalable system architectures, inadequate reporting of real-time latency, and limited command set diversity. The proposed system design addresses each of these gaps through deliberate and traceable architectural decisions — public dataset usage, a fully modular five-stage pipeline, SVM-based lightweight classification, PyBullet simulation integration, confidence-threshold command filtering, and a targeted 4–5 motor imagery command set.

The measurable performance targets — classification accuracy $\geq 75\%$, command success rate $\geq 80\%$, and system latency ≤ 500 ms — are grounded in and consistent with the best-performing lightweight systems documented in the reviewed literature, providing a realistic yet rigorous framework for evaluation.

Future work will focus on three development directions: transitioning from simulation to physical robotic hardware; exploring transfer learning for cross-subject generalisation to reduce per-user calibration burden; and investigating consumer-grade EEG headsets as a pathway toward truly affordable and portable real-world deployment. Together, these directions position the proposed

system as a scalable, accessible, and academically rigorous contribution to the field of assistive BCI technology within next-generation smart healthcare and rehabilitation environments.

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